Project Design Phase
Problem – Solution Fit Template

Date	2 March 2026
Team ID	MBA-041
Project Name	Heart Disease Analysis using Tableau
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem–Solution Fit explains how the proposed system effectively addresses the identified problem and meets user needs. In this project, the primary problem is the difficulty in extracting meaningful insights from large-scale heart disease datasets. Traditional data analysis methods often rely on static reports, which limit the ability to explore data dynamically and identify patterns quickly.

Healthcare professionals, policymakers, and individuals require a solution that simplifies complex medical data and presents it in an intuitive and interactive format. Without proper visualization tools, identifying correlations between lifestyle factors and heart disease risk becomes time consuming and inefficient. This gap highlights the need for a data-driven solution capable of transforming raw health data into actionable insights.

The proposed Tableau-based analytics system fits this need by providing interactive dashboards and visual stories that enable users to explore data in real time. Tableau's ability to handle large datasets and generate dynamic visualizations ensures that users can filter, compare, and analyze multiple variables simultaneously. This directly addresses the problem of limited data interpretability.

Additionally, integrating the dataset into a structured database ensures efficient storage and retrieval, while SQL operations allow preprocessing and transformation of data to suit analytical needs. The combination of database management and visualization tools creates a seamless workflow from data acquisition to insight generation.

Overall, the solution aligns strongly with the problem by improving accessibility to insights, enhancing decision-making capabilities, and enabling proactive measures for heart disease prevention. The system not only solves the technical challenge of data analysis but also delivers tangible value to stakeholders by supporting informed actions.

✅ 1. Problem Statement

What is the customer struggling with?

Stakeholders in the real estate sector—such as analysts, marketing teams, and executives—face challenges in:

- Identifying which property features influence pricing trends
- Understanding how renovations affect buyer interest and price
- Making strategic decisions without data-backed insights
- Communicating property trends effectively to clients or investors

💡 2. Existing Behavior / Situation

How are they currently addressing the problem?

- Based on spreadsheets and static reports with limited visual context
- Heavily reliant on manual analysis with potential data misinterpretation
- Missing comprehensive dashboards for interactive exploration
- Not leveraging full potential of historical or feature-specific data

🎯 3. Desired Outcome / Ideal Scenario

What would a better world look like for them?

- Ability to visually explore and interpret housing trends easily
- Understand how each feature (renovation, age, rooms/floors) impacts sale price
- Make informed and fast decisions using an interactive Tableau dashboard
- Identify actionable trends and optimize pricing strategies

🛠️ 4. Our Solution

How does your solution address the problem effectively?

An interactive Tableau dashboard that:

- Provides overall data summaries with key KPIs (Scenario 1)
- Visualizes sales trends based on years since renovation (Scenario 2)
- Breaks down age distribution by renovation status (Scenario 3)
- Analyzes house features like bathrooms, bedrooms, and floors by age (Scenario 4)
- Offers drill-down capabilities and filters for customized insights
- Can be embedded in a web app using Flask for broader accessibility

🚀 5. Why It Works

How does it align with customer behavior and needs?

- Real estate teams already work with data but lack intuitive tools → Tableau adds clarity
- Renovations and house features are top decision factors → directly visualized
- Users prefer visually rich, interactive reports over static spreadsheets
- Immediate insights help close decisions faster, leading to business growth

📣 6. Marketing / Communication Strategy

What messaging and touch-points help adoption?

- Messaging Angle: "Turn your housing data into decisions"
- Emotional Trigger: Solve the frustration of slow, unclear analysis
- Rational Trigger: Improve accuracy and speed of market evaluations
- Touch-points: Demo sessions, tutorials, integration in existing dashboards
- Performance-based dashboards shared with stakeholders